

a complete, **step-by-step solved numerical example** of a small multilayer perceptron (MLP) — forward pass, backprop derivatives, weight updates, and the new error after one training update.

Network & initial values

- Architecture: 2 inputs → 2 hidden neurons → 2 output neurons
- Activation: sigmoid $\sigma(z) = 1/(1 + e^{-z})$
- Inputs: $x_1 = 0.05$, $x_2 = 0.10$
- Targets: $t_1 = 0.01$, $t_2 = 0.99$
- Initial weights and biases:
 - Input → Hidden: $w_1 = 0.15$, $w_2 = 0.20$, $w_3 = 0.25$, $w_4 = 0.30$
 - Hidden → Output: $w_5 = 0.40$, $w_6 = 0.45$, $w_7 = 0.50$, $w_8 = 0.55$
 - Hidden bias $b_1 = 0.35$, Output bias $b_2 = 0.60$
- Learning rate: $\eta = 0.5$
- Loss: sum of squared errors $E = \frac{1}{2}[(t_1 - o_1)^2 + (t_2 - o_2)^2]$

1) Forward pass (initial)

Compute hidden layer nets and activations:

- $h1_{in} = x_1w_1 + x_2w_2 + b_1 = 0.05 \cdot 0.15 + 0.10 \cdot 0.20 + 0.35 = 0.3775$
- $h2_{in} = x_1w_3 + x_2w_4 + b_1 = 0.05 \cdot 0.25 + 0.10 \cdot 0.30 + 0.35 = 0.3925$

Hidden activations (sigmoid):

- $h_1 = \sigma(0.3775) = 0.5932699921$
- $h_2 = \sigma(0.3925) = 0.5968843783$

Output nets and activations:

- $o1_{in} = h_1w_5 + h_2w_6 + b_2 = 0.5932699921 \cdot 0.40 + 0.5968843783 \cdot 0.45 + 0.60 = 1.1059059671$
- $o2_{in} = h_1w_7 + h_2w_8 + b_2 = 0.5932699921 \cdot 0.50 + 0.5968843783 \cdot 0.55 + 0.60 = 1.2249214041$
- $o_1 = \sigma(1.1059059671) = 0.7513650696$
- $o_2 = \sigma(1.2249214041) = 0.7729284653$

Initial total error:

$$E_{\text{initial}} = \frac{1}{2}[(0.01 - 0.7513650696)^2 + (0.99 - 0.7729284653)^2] = 0.2983711088$$

2) Backpropagation — gradients

We compute gradients of the loss w.r.t. each weight (chain rule). Useful derivatives:

- For output unit k : $\frac{\partial E}{\partial o_k} = o_k - t_k$
- Sigmoid derivative: $\sigma'(y) = y(1 - y)$ for activation value y

Output-layer weight gradients

- $\delta_{o1} = (o_1 - t_1) \cdot \sigma'(o_1) = (0.7513650696 - 0.01) \cdot (0.7513650696(1 - 0.7513650696))$

This yields gradients for w_5, w_6 :

- $\frac{\partial E}{\partial w_5} = \delta_{o1} \cdot h_1 = 0.08216704056$
- $\frac{\partial E}{\partial w_6} = \delta_{o1} \cdot h_2 = 0.08266762785$
- $\delta_{o2} = (o_2 - t_2) \cdot \sigma'(o_2)$
 - $\frac{\partial E}{\partial w_7} = \delta_{o2} \cdot h_1 = -0.02260254048$
 - $\frac{\partial E}{\partial w_8} = \delta_{o2} \cdot h_2 = -0.02274024222$

Bias gradient for output bias b_2 :

- $\frac{\partial E}{\partial b_2} = \delta_{o1} + \delta_{o2} = 0.1004003251$

Hidden-layer weight gradients (via backprop)

Backprop errors to hidden units:

- $\delta_{h1} = (\delta_{o1}w_5 + \delta_{o2}w_7) \cdot \sigma'(h_1)$
- $\delta_{h2} = (\delta_{o1}w_6 + \delta_{o2}w_8) \cdot \sigma'(h_2)$

Resulting gradients:

- $\frac{\partial E}{\partial w_1} = \delta_{h1} \cdot x_1 = 0.0004385677345$
- $\frac{\partial E}{\partial w_2} = \delta_{h1} \cdot x_2 = 0.0008771354689$
- $\frac{\partial E}{\partial w_3} = \delta_{h2} \cdot x_1 = 0.0004977127353$
- $\frac{\partial E}{\partial w_4} = \delta_{h2} \cdot x_2 = 0.0009954254705$

Bias gradient for hidden bias b_1 :

- $\frac{\partial E}{\partial b_1} = \delta_{h1} + \delta_{h2} = 0.01872560939$

3) Weight updates (gradient descent)

Update rule: $w \leftarrow w - \eta \frac{\partial E}{\partial w}$ with $\eta = 0.5$

Updated weights (rounded shown):

- $w_1 \leftarrow 0.15 - 0.5 \cdot 0.0004385677345 = \mathbf{0.1497807161}$
- $w_2 \leftarrow 0.20 - 0.5 \cdot 0.0008771354689 = \mathbf{0.1995614323}$
- $w_3 \leftarrow 0.25 - 0.5 \cdot 0.0004977127353 = \mathbf{0.2497511436}$
- $w_4 \leftarrow 0.30 - 0.5 \cdot 0.0009954254705 = \mathbf{0.2995022873}$

- $w_5 \leftarrow 0.40 - 0.5 \cdot 0.08216704056 = \mathbf{0.3589164797}$
- $w_6 \leftarrow 0.45 - 0.5 \cdot 0.08266762785 = \mathbf{0.4086661861}$
- $w_7 \leftarrow 0.50 - 0.5 \cdot (-0.02260254048) = \mathbf{0.5113012702}$
- $w_8 \leftarrow 0.55 - 0.5 \cdot (-0.02274024222) = \mathbf{0.5613701211}$

Bias updates:

- $b_1 \leftarrow 0.35 - 0.5 \cdot 0.01872560939 = \mathbf{0.3406371953}$
- $b_2 \leftarrow 0.60 - 0.5 \cdot 0.1004003251 = \mathbf{0.5497998374}$

4) Forward pass after update (one training step)

Compute outputs with updated weights:

- New hidden activations: $h_1 = 0.5909955313$, $h_2 = 0.5946145364$
- New outputs: $o_1 = 0.7320241671$, $o_2 = 0.7659846534$

New total error:

$$E_{\text{new}} = \frac{1}{2}[(0.01 - 0.7320241671)^2 + (0.99 - 0.7659846534)^2] = 0.2857508867$$

Error decreased from 0.2983711088 $\rightarrow$ 0.2857508867 after a single update.

Summary / takeaway

- This shows one full forward+backprop update for a tiny 2-2-2 MLP with sigmoid activations.
- You can repeat this process (many iterations / epochs) to keep reducing error.
- The key steps are:
 1. Forward pass to compute activations and loss.
 2. Backprop to compute gradients $\partial E / \partial w$.
 3. Update weights by gradient descent.
 4. Repeat.